Associations between cadmium exposure and neurocognitive test scores in a cross-sectional study of US adults

Background Low-level environmental cadmium exposure and neurotoxicity has not been well studied in adults. Our goal was to evaluate associations between neurocognitive exam scores and a biomarker of cumulative cadmium exposure among adults in the Third National Health and Nutrition Examination Survey (NHANES III). Methods NHANES III is a nationally representative cross-sectional survey of the U.S. population conducted between 1988 and 1994. We analyzed data from a subset of participants, age 20–59, who participated in a computer-based neurocognitive evaluation. There were four outcome measures: the Simple Reaction Time Test (SRTT: visual motor speed), the Symbol Digit Substitution Test (SDST: attention/perception), the Serial Digit Learning Test (SDLT) trials-to-criterion, and the SDLT total-error-score (SDLT-tests: learning recall/short-term memory). We fit multivariable-adjusted models to estimate associations between urinary cadmium concentrations and test scores. Results 5662 participants underwent neurocognitive screening, and 5572 (98%) of these had a urinary cadmium level available. Prior to multivariable-adjustment, higher urinary cadmium concentration was associated with worse performance in each of the 4 outcomes. After multivariable-adjustment most of these relationships were not significant, and age was the most influential variable in reducing the association magnitudes. However among never-smokers with no known occupational cadmium exposure the relationship between urinary cadmium and SDST score (attention/perception) was significant: a 1 μg/L increase in urinary cadmium corresponded to a 1.93% (95%CI: 0.05, 3.81) decrement in performance. Conclusions These results suggest that higher cumulative cadmium exposure in adults may be related to subtly decreased performance in tasks requiring attention and perception, particularly among those adults whose cadmium exposure is primarily though diet (no smoking or work based cadmium exposure). This association was observed among exposure levels that have been considered to be without adverse effects and these levels are common in U.S. adults. Thus further research into the potential neurocognitive effects of cadmium exposure is warranted. Because cumulative cadmium exposure may mediate some of the effects of age and smoking on cognition, adjusting for these variables may result in the underestimation of associations with cumulative cadmium exposure. Prospective studies that include never-smokers and non-occupationally exposed individuals are needed to clarify these issues.